

CRM Sync — UI Component & ID Registry

Canonical naming + delivery model for the storefront UI system (nav, footer, cart, login, search) across the store and `crm-sync.dev` — **one addressable `crm-` namespace** so every surface, collection, and bundle lines up. Live endpoints marked; planned ones tracked.

Canonical naming + delivery model for the storefront UI system (nav, footer, cart, login, search) across `design-sync.myshopify.com` → `crm-sync.dev`. One addressable `crm-` namespace so every surface, collection, and bundle lines up.

Status: working reference. Endpoints marked ✓ are live; ⌚ are planned.

1. DELIVERY MODEL — THREE APP BUNDLES, LOADED FROM THE CF WORKER

CRM, PIM, and Design each ship as a **separate component bundle** served from the Cloudflare Worker's `/embed/*` endpoints. Updating a bundle = a worker deploy; the storefront never re-pastes code. (Layer 4 of the Higher-Order Stack.)

BUNDLE	LOADS	ENDPOINT (WORKER)	PLACEMENT
Stack (Layer 1)	GA4 (Consent-Mode-v2, id from <code>/stack/config</code>) + Ulkit/GSAP/petite-vue per-need	<code>/embed/stack-loader.js</code> ✓	helmet (<code><head></code>)
CRM	<code><crm-brand-studio></code> , <code><crm-brand></code> , <code><crm-sync></code> (event bus)	<code>/embed/crm-elements.js</code> ✓	defer
PIM	<code><pim-*></code> + <code>crmPim.resolve()</code> GID⇌Webflow⇌Xano	<code>/embed/pim-elements.js</code> ✓	defer
Design	brand theme / tokens (Layer 2 look)	<code>/brand/<slug>/theme.css</code> ✓	helmet
Docs modal	<code>window.crmDocsModal</code> (docs utility)	Pages <code>docs-nav.js</code> (data-modal-only)	defer

Bundles are **served from the worker**, so an update ships with `wrangler deploy` — the storefront never re-pastes code. The Stack loader is `window.crmStack` (`data-stack="uikit"`, `data-worker`, `data-shop`; GA4 pulled from `/stack/config` per-shop, MP secret server-side). The Brand Designer loads the SAME loader with `data-stack="uikit,petite-vue,gsap"`.

Placement rule: the Nav loads in the helmet (head — no FOUC, paints first). The Footer loads deferred (`defer`) at end of body and carries the Shopify / CRM Web Components for **Login + Cart**.

Header nav (shopify-uikit-nav.liquid)

ELEMENT	ID	CLASS	NOTES
Nav wrapper / mount	—	.crm-nav	petite-vue mount root
Search	#crm-nav-search	.crm-nav-search	UIKit search icon → routes.search_url
Cart	#crm-nav-cart	.crm-nav-cart	UIKit cart icon + [data-cart-count] badge
Login	#crm-nav-login	.crm-nav-login	→ routes.account_login_url
CTA	—	.crm-nav-cta	primary button
Offcanvas (mobile)	#crm-nav-offcanvas	—	UIKit offcanvas
Mobile search / cart / login	#crm-nav-search-m · #crm-nav-cart-m · #crm-nav-login-m		inside offcanvas

Footer (shopify-uikit-footer.liquid ⌚)

ELEMENT	ID	NOTES
Footer wrapper	#crm-footer	render root
Login (web component)	#crm-nav-login	shared login id — <crm-login> / Shopify Web Component
Cart (web component)	#crm-nav-cart	shared cart id — <crm-cart>
Footer nav mount	#crm-footer-nav	from GET /nav?menu=footer

Cart and Login share **one id each** across nav + footer so a single cart/login component instance binds regardless of which surface triggered it.

Collections (Webflow → sync → worker)

COLLECTION	LIST ELEMENT ID	ITEM / LINK CLASS	MENU KEY
Nav Menu (header)	#crm-nav-collection	.crm-nav-item / .crm-nav-link	main
Footer Menu	#crm-footer-collection	.crm-footer-item / .crm-footer-link	footer
Tags → Category	#crm-category-collection	.crm-category-item	(category tables)

Collection field slugs (what the sync reads): title , url , active , locale (+ optional order , group).

3. DATA ENDPOINTS (WORKER)

ENDPOINT	METHOD	AUTH	PURPOSE
/stack/config?shop=	GET	public	GA4 measurement ID (public subset only) ✓
/nav?shop=&menu=&locale=	GET	public	named menu (main / footer), localized ✓
/nav?shop=&menu=&locale=	POST	admin/tenant	write a menu (Webflow sync / config app) ✓
/flow/campaign	POST	admin/tenant	Shopify Flow action → GA4 Smart Bidding (consent-gated, revenue-weighted) ✓
/categories?shop=&locale=&kind=	GET/POST	public / admin	Tags as Category Collection ✓ (kind filter; mirrors /nav)

`menu` resolution: `nav_menus[menu].i18n[locale]` → `[lang]` → `.items`. Each locale is a separately-editable instance (English + globalized variants).

4. TAGS = CATEGORY COLLECTION TABLES

Tags are modeled as a **Category Collection** – the same collection→sync→worker pattern as nav, backed by the category tables (`channels(201)` / `channel_membership(202)` / `category_pivot(203)`). A Webflow "Category" collection (`#crm-category-collection`) syncs to `/categories` , and category-driven UI (filters, tag chips, audience membership) reads from it – one authoring surface, localized, projected to every surface.

5. WHERE EACH PIECE GOES (PASTE MAP)

PIECE	THEME LOCATION	LOADING
Stack loader + Ulkit CSS + nav <code><style></code>	<code><head></code> (helmet)	blocking CSS, <code>async</code> JS
Nav markup	top of body / section	server-rendered Liquid
Footer markup + CRM/PIM elements	before <code></body></code>	<code>defer</code>
Brand theme.css (Design)	<code><head></code>	blocking

Long-term: the worker's Theme App Extension **app embed** injects the helmet + deferred bundles, so there's no manual theme editing.

Four axes are modeled as **Triggers** whose **Actions** execute in **server-side functions** (worker functions / Shopify Functions) – never in theme JS. They ride the consent-aware event bus (Tier A), are **fail-closed**, and are audit-logged.

AXIS	TRIGGER (FIRES ON...)	ACTION (SERVER-SIDE)	SSR FUNCTION / SEAM
Brand	brand/theme selected or published	re-theme — emit <code>theme.css</code> / tokens for the surface	<code>/brand/<slug>/theme.css</code> · Design bundle
QA / PROD	promote between realms (Stage → Prod → Deploy-Live)	gate + swap env-scoped config/creds; fail-closed if connections/approvals incomplete	<code>/brand/<slug>/promote</code> · env-label config
Persona	role / entitlement change (Designer, QA, Release Eng)	cap check at the data plane ; hide-by-cap in UI, enforce in worker	<code>entitlements(190)</code> · <code>hasCap()</code> / <code>userIsDesignerForBrand()</code>
Event	consent change · cart · AP2 mandate · A2A delegation	run the handler (project to GA4/ESB, gate the cart, settle)	consent-gated <code>dataLayer</code> bus → worker/Shopify Functions

The rule: a Trigger is a condition on the bus; its Action is a **deterministic server-side function** (input-bounded, side-effect-scoped), re-checked per loop. Presentation (nav/footer/components) only *reflects* the result — it never decides. This is why the same trigger holds whether a human or an agent fired it.

Backs onto: release personas / separation-of-duties, the brand env realms (Stage-Agency / Prod-Agency / Deploy-Live), and `entitlement_changes(193)` / AP2 `agentic_checkout` on the event bus.

Invariant: every Shopify **GID** (`gid://shopify/Product/...`, Collection, Customer, Order, ...) has a **same-name item in Webflow**, joined to a **Xano row** as the durable source of truth. One entity, three representations, one natural key.

```

SHOPIFY gid://...  $\rightleftharpoons$  XANO row (SoR, natural key + gid)  $\rightleftharpoons$  WEBFLOW item (same name/slug)

      ▲                               ▲
      |                               |
      | sockets (CF Worker)          |
      |                               |
      └── AI / CF / Xano orchestration ─┘

```

Rules (from the ORM discipline in `PROCESS-MANAGEMENT-DATA-LAYER.md`):

- **Stable natural key** (SKU/handle/name), not a platform id, is the join — the item survives being re-created in any one platform. The GID is carried as an attribute, not the key.
- **Xano is the source of truth**; Shopify and Webflow are projections. Data flows **out of Xano**; no channel is the only place a fact exists.
- **One writer per field** (price $\rightarrow$ Xano, published-state $\rightarrow$ Webflow, fulfillment $\rightarrow$ Shopify); versioned upserts, idempotent on `(natural_key, source)`.
- **Sockets = CF Worker** owns the mapping + idempotency + conflict resolution; **AI** orchestrates match/normalize/enrich; **Xano** persists. Same seam the nav / footer / category collections ride — the collection items are just GID-keyed rows projected to each surface.

This makes every entity **agent-addressable and consent-qualified** end to end: an AI agent resolves a Shopify GID $\rightarrow$ Xano row $\rightarrow$ Webflow item (and back) through one worker socket, under caps + consent, per loop.

The **Ulkit / semantic wrapper** is what makes every `#crm-` id machine-legible. Because `uk-*` is BEM-namespaced (conflict-free by construction) it supplies the semantic skeleton that utility CSS can't; on top of it we attach a **role + attribute pattern keyed to the id** so screen readers *and* answer engines parse the same structure.

Pattern per addressable element:

LAYER	CARRIES	EXAMPLE (<code>#CRM-NAV-CART</code>)
id	the address	<code>id="crm-nav-cart"</code>
ARIA role / label	a11y semantics (WCAG)	<code>aria-label="Cart"</code>
<code>data-crm-role</code>	machine role for AEO / agents	<code>data-crm-role="cart"</code>
<code>data-crm-region</code>	landmark on the wrapper	<code><nav ... data-crm-region="primary-nav"></code>
Ulkit BEM class	component structure	<code>.uk-navbar-item</code>

Applied on the nav today: `data-crm-region="primary-nav"` on the `<nav>` landmark; `data-crm-role="search|cart|login"` on the controls (+ native `role` / `aria-label`). Same pattern extends to footer, category chips, and every `<crm-*>` component – one wrapper, three readers (browser, screen reader, answer engine / agent).

Scores against the a11y (WCAG) + machine-index (AEO) harness – the wrapper is how we keep both high without hand-tuning each surface.